TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	88.6 ± 1.02
Correlation Stability	93.5 ± 0.67
Distribution Consistency	93.4 ± 0.92
Entropy Preservation	89.9 ± 0.70
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	100.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	0.0 ± 0.00
Feature Accuracy	94.5 ± 1.36
Novelty Score	0.0 ± 0.00

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	88.6 ± 1.02
Correlation Stability (*2)	93.5 ± 0.67
Distribution Consistency (*3)	93.4 ± 0.92
Entropy Preservation (*4)	89.9 ± 0.70
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	100.0 ± 0.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	0.0 ± 0.00
Feature Accuracy (*10)	94.5 ± 1.36
Novelty Score (*11)	0.0 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.727272727272727 3	N/A	0.727272727272727 3	0	
F1	0.590668774601382 5	N/A	0.590668774601382 5	0	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.746753246753246 8	N/A	0.746103896103896	-0.00064935064935065 51	
F1	0.618167796917797	N/A	0.616694112707270 7	-0.00147368421052631 58	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.711038961038961	N/A	0.709740259740259 8	-0.00129870129870129 91	
F1	0.570057748495880 6	N/A	0.572952090071979 3	0.00289434157609876 9	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.756493506493506 4	N/A	0.756493506493506 4	0	
F1	0.573368073405794 2	N/A	0.573368073405794 2	0	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.772727272727272 7	N/A	0.772727272727272 7	0	
F1	0.627868128746641 9	N/A	0.627868128746641 9	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	N/A	0.7493506493506492	0	
F1	0.6076604413342895	N/A	0.6076604413342895	0	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7636363636363636	N/A	0.7636363636363636	0	
F1	0.6399852354635025	N/A	0.6399852354635025	0	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7545454545454545	N/A	0.7545454545454545	0	
F1	0.5998259463389416	N/A	0.5998259463389416	0	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6551948051948052	N/A	0.662987012987013	0.007792207792207772	
F1	0.4915419652387786	N/A	0.4979169136655438	0.006374948426765159	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7558441558441558	N/A	0.7538961038961038	-0.0019480519480519302	
F1	0.6124836913044565	N/A	0.6146518065511126	0.0021681152466561937	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	127.0	88.0	11.0	155.0	34.5	0.598	28.0	0.0	1.0
2	1.0	130.0	70.0	13.0	105.0	25.9	0.472	22.0	0.0	1.0
3	1.0	87.0	78.0	27.0	32.0	34.6	0.101	22.0	0.0	1.0
4	5.0	144.0	82.0	26.0	285.0	32.0	0.452	58.0	1.0	1.0
5	11.0	138.0	74.0	26.0	144.0	36.1	0.557	50.0	1.0	1.0
6	3.0	123.0	100.0	35.0	240.0	57.3	0.88	22.0	0.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	3.0	99.0	62.0	19.0	74.0	21.8	0.279	26.0	0.0	2.0
2	8.0	126.0	74.0	38.0	75.0	25.9	0.162	39.0	0.0	2.0
3	3.0	132.0	80.0	0.0	0.0	34.4	0.402	44.0	1.0	2.0
4	12.0	100.0	84.0	33.0	105.0	30.0	0.488	46.0	0.0	2.0
5	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0	2.0
6	3.0	111.0	58.0	31.0	44.0	29.5	0.43	22.0	0.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	8.0	186.0	90.0	35.0	225.0	34.5	0.423	37.0	1.0	3.0
2	1.0	108.0	60.0	46.0	178.0	35.5	0.415	24.0	0.0	3.0
3	4.0	99.0	72.0	17.0	0.0	25.6	0.294	28.0	0.0	3.0
4	5.0	88.0	78.0	30.0	0.0	27.6	0.258	37.0	0.0	3.0
5	1.0	116.0	78.0	29.0	180.0	36.1	0.496	25.0	0.0	3.0
6	0.0	91.0	80.0	0.0	0.0	32.4	0.601	27.0	0.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

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B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	3.0	120.0	70.0	30.0	135.0	42.9	0.452	30.0	0.0	4.0
2	7.0	160.0	54.0	32.0	175.0	30.5	0.588	39.0	1.0	4.0
3	1.0	172.0	68.0	49.0	579.0	42.4	0.702	28.0	1.0	4.0
4	1.0	128.0	88.0	39.0	110.0	36.5	1.057	37.0	1.0	4.0
5	1.0	125.0	50.0	40.0	167.0	33.3	0.962	28.0	1.0	4.0
6	10.0	115.0	0.0	0.0	0.0	0.0	0.261	30.0	1.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	125.0	78.0	31.0	0.0	27.6	0.565	49.0	1.0	5.0
2	6.0	103.0	66.0	0.0	0.0	24.3	0.249	29.0	0.0	5.0
3	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	5.0
4	2.0	122.0	52.0	43.0	158.0	36.2	0.816	28.0	0.0	5.0
5	4.0	115.0	72.0	0.0	0.0	28.9	0.376	46.0	1.0	5.0
6	5.0	123.0	74.0	40.0	77.0	34.1	0.269	28.0	0.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	154.0	62.0	31.0	284.0	32.8	0.237	23.0	0.0	6.0
2	1.0	108.0	60.0	46.0	178.0	35.5	0.415	24.0	0.0	6.0
3	1.0	125.0	70.0	24.0	110.0	24.3	0.221	25.0	0.0	6.0
4	0.0	101.0	62.0	0.0	0.0	21.9	0.336	25.0	0.0	6.0
5	3.0	142.0	80.0	15.0	0.0	32.4	0.2	63.0	0.0	6.0
6	7.0	103.0	66.0	32.0	0.0	39.1	0.344	31.0	1.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

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B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	115.0	72.0	0.0	0.0	28.9	0.376	46.0	1.0	7.0
2	4.0	129.0	60.0	12.0	231.0	27.5	0.527	31.0	0.0	7.0
3	3.0	90.0	78.0	0.0	0.0	42.7	0.559	21.0	0.0	7.0
4	1.0	71.0	78.0	50.0	45.0	33.2	0.422	21.0	0.0	7.0
5	9.0	124.0	70.0	33.0	402.0	35.4	0.282	34.0	0.0	7.0
6	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	172.0	68.0	49.0	579.0	42.4	0.702	28.0	1.0	8.0
2	5.0	139.0	64.0	35.0	140.0	28.6	0.411	26.0	0.0	8.0
3	2.0	92.0	62.0	28.0	0.0	31.6	0.13	24.0	0.0	8.0
4	4.0	183.0	0.0	0.0	0.0	28.4	0.212	36.0	1.0	8.0
5	2.0	157.0	74.0	35.0	440.0	39.4	0.134	30.0	0.0	8.0
6	4.0	146.0	78.0	0.0	0.0	38.5	0.52	67.0	1.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	2.0	197.0	70.0	99.0	0.0	34.7	0.575	62.0	1.0	9.0
2	0.0	102.0	86.0	17.0	105.0	29.3	0.695	27.0	0.0	9.0
3	5.0	158.0	70.0	0.0	0.0	29.8	0.207	63.0	0.0	9.0
4	2.0	91.0	62.0	0.0	0.0	27.3	0.525	22.0	0.0	9.0
5	9.0	140.0	94.0	0.0	0.0	32.7	0.734	45.0	1.0	9.0
6	9.0	123.0	70.0	44.0	94.0	33.1	0.374	40.0	0.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

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B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	131.0	0.0	0.0	0.0	43.2	0.27	26.0	1.0	10.0
2	9.0	165.0	88.0	0.0	0.0	30.4	0.302	49.0	1.0	10.0
3	0.0	132.0	78.0	0.0	0.0	32.4	0.393	21.0	0.0	10.0
4	9.0	164.0	78.0	0.0	0.0	32.8	0.148	45.0	1.0	10.0
5	4.0	85.0	58.0	22.0	49.0	27.8	0.306	28.0	0.0	10.0
6	1.0	143.0	84.0	23.0	310.0	42.4	1.076	22.0	0.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan